Renewable energy consumption and Co2 emissions from power industry

RAMRACHAI MARMA - 11951184

***Abstract*—The relationship between renewable energy consumption and carbon dioxide (CO2) emissions is critical to understanding the transition toward sustainable energy systems. This study examines the impact of renewable energy consumption on reducing CO2 emissions using two datasets sourced from the World Bank, which detail renewable energy consumption (percent of total final energy consumption) and CO2 emissions (metric tons per capita). By analyzing temporal and geographical trends, the study identifies correlations and explores the potential of renewable energy as a mitigation strategy for climate change.**

I. Introduction

The escalating threat of climate change has made reducing carbon dioxide (CO2) emissions a global priority. The energy sector, which is predominantly powered by fossil fuels, remains a major contributor to greenhouse gas emissions. To address this challenge, a transition toward renewable energy sources is increasingly seen as a key solution. Renewable energy sources, such as solar, wind, and hydropower, offer the dual benefit of reducing reliance on fossil fuels and mitigating environmental damage.

This report explores a fundamental question: Does an increase in renewable energy consumption reduce CO2 emissions? To answer this, two datasets from the World Bank are analyzed—one providing data on renewable energy consumption as a percentage of total final energy consumption and the other tracking CO2 emissions per capita across various regions and time periods. By examining these trends, the report seeks to assess whether the adoption of renewable energy is an effective strategy to reduce CO2 emissions.

II. Data Source

A. *Data Usage*

For the analysis, we utilized two datasets from the World Bank: renewable energy consumption and CO2 emissions. The dataset was structured and processed using an automated ETL pipeline to ensure accuracy and consistency for analysis.

Renewable Energy Consumption Dataset:

This dataset provides annual data on the percentage of renewable energy consumption as part of total final energy consumption across countries and regions. The dataset spans multiple years, offering insights into the growth of renewable energy usage and its distribution globally.

CO2 Emissions Dataset:

This dataset contains annual records of carbon dioxide (CO2) emissions, measured in metric tons per capita, across countries and regions. It helps capture the environmental impact of energy consumption patterns and other economic activities. The combination of these datasets allows a comprehensive examination of the relationship between renewable energy use and CO2 emissions over time.

Identify applicable funding agency here. If none, delete this.

B. *Data Pipeline*

The data pipeline ensures that both datasets are cleaned, transformed, and prepared for analysis. The final dataset is complete with no significant missing values and no irrelevant features, providing high-quality input for reliable insights.

Temporal Alignment: Data from both sources was aligned to ensure consistency in the timeframes under analysis.

Feature Engineering: New features, such as regional aggregates and year-over-year changes, were added to enhance the analytical depth of the dataset.

Normalization and Scaling: The features, such as renewable energy consumption and CO2 emissions, were normalized to ensure comparability and eliminate biases caused by differences in numerical ranges. Without scaling, features with larger ranges could skew the analysis and lead to misleading interpretations.

Data Quality Checks: The pipeline performed checks to remove duplicate records, handle outliers, and ensure data integrity.

C. *Data Structure*

Both datasets are structured with temporal, categorical, and continuous features to ensure maximum coverage of relevant factors:

Renewable Energy Consumption Dataset:

Temporal Features: Year of observation (e.g., 2000-2020). Categorical Features: Country, Region. Continuous Features: Renewable energy consumption percent of total final energy consumption.

Country Name	Country Code	Indicator Name	Year	Value
Bermuda	BMU	Renewable energy consumption (% of total final...	2019	0.8
Costa Rica	CRI	Renewable energy consumption (% of total final...	2019	33.4
Cuba	CUB	Renewable energy consumption (% of total final...	2019	23.1
Bermuda	BMU	Renewable energy consumption (% of total final...	2020	0.9
Costa Rica	CRI	Renewable energy consumption (% of total final...	2020	36.4
Cuba	CUB	Renewable energy consumption (% of total final...	2020	24.5
Bermuda	BMU	Renewable energy consumption (% of total final...	2021	0.9
Costa Rica	CRI	Renewable energy consumption (% of total final...	2021	34.2
Cuba	CUB	Renewable energy consumption (% of total final...	2021	20.9

Fig. 1. Renewable energy consumption data for Bermuda, Cuba, and Costa Rica.

CO2 Emissions Dataset:

Temporal Features: Year of observation.

Categorical Features: Country, Region.

Continuous Features: CO2 emissions (metric tons per capita).

By combining these structured datasets, the analysis explores the correlation between renewable energy adoption and CO2 emissions reduction, uncovering key trends and actionable insights.

Country Name	Country Code	Indicator Name	Year	Value
Bermuda	BMU	Carbon dioxide (CO2) emissions from Power Indu...	2019	0.1760
Costa Rica	CRI	Carbon dioxide (CO2) emissions from Power Indu...	2019	0.0651
Cuba	CUB	Carbon dioxide (CO2) emissions from Power Indu...	2019	13.9231
Bermuda	BMU	Carbon dioxide (CO2) emissions from Power Indu...	2020	0.1621
Costa Rica	CRI	Carbon dioxide (CO2) emissions from Power Indu...	2020	0.0208
Cuba	CUB	Carbon dioxide (CO2) emissions from Power Indu...	2020	13.8200
Bermuda	BMU	Carbon dioxide (CO2) emissions from Power Indu...	2021	0.1562
Costa Rica	CRI	Carbon dioxide (CO2) emissions from Power Indu...	2021	0.0237
Cuba	CUB	Carbon dioxide (CO2) emissions from Power Indu...	2021	13.5938

Fig. 2. CO2 emissions data for Bermuda, Cuba, and Costa Rica.

D. *License*

Both datasets are provided under the Creative Commons Attribution 4.0 International (CC BY 4.0) license, which allows for sharing and adapting the data for any purpose, as long as proper credit is given. To comply with the licensing requirements, I will ensure the World Bank is appropriately credited as the source in all reports, presentations, or publications resulting from this analysis. For detailed information about the license, please visit the Creative Commons Attribution 4.0 International License page at https://creativecommons.org/licenses/by/4.0/.

III. ANALYSIS

To understand how renewable energy consumption impacts CO2 emissions, we performed several data visualization techniques (exploratory data analysis) and correlation analysis on the dataset. By examining these analyses across various countries and timeframes, we identified significant relationships between renewable energy adoption and CO2 emission patterns. These findings provide valuable insights into the role of renewable energy in mitigating climate change and highlight opportunities to promote sustainable energy practices for reducing global carbon footprints.

A. *Exploratory Data Analysis (EDA)*

From the two graphs, we analyze the trends in renewable energy consumption and CO2 emissions over time (2015–2021) for the countries: Bermuda, Costa Rica, and Cuba.

Renewable Energy Consumption

Costa Rica consistently leads in renewable energy consumption, maintaining values between 33 percent and 38 percent. There is a slight dip in 2016 but a recovery in 2019–2020, followed by a small decline in 2021. Cuba shows moderate renewable energy usage, fluctuating between 22 percent and 25 percent. A noticeable peak is seen in 2020, followed by a drop in 2021. Bermuda has very low renewable energy consumption throughout the years, staying below 1 percent, with minimal variation over time.

Key insights: Costa Rica exhibits a strong commitment to renewable energy, while Bermuda lags significantly. Cuba shows slight improvements, but progress is inconsistent.

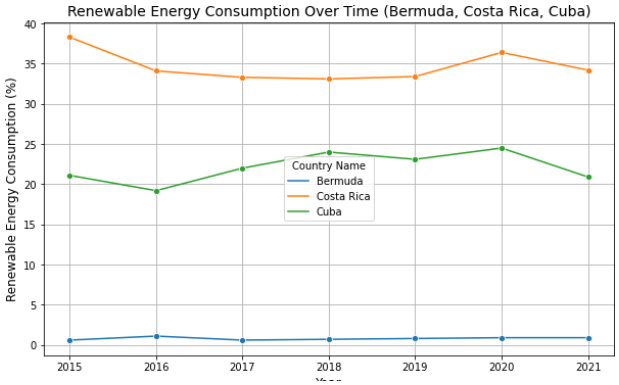


Fig. 3. Renewable energy consumption graph for Bermuda, Cuba, and Costa Rica.

CO2 Emissions

Cuba shows significantly higher CO2 emissions compared to Costa Rica and Bermuda, with emissions decreasing after 2016 from around 16 percent to 13.5 percent. Costa Rica and Bermuda have very low CO2 emissions, with minimal variations over the years.

Key Insights:

Cuba's emissions decreased after 2016, likely due to a reduction in fossil fuel dependence or other mitigation strategies. Costa Rica and Bermuda maintain low CO2 emissions, consistent with their relatively low industrial emissions and increased adoption of renewable energy (in the case of Costa Rica).

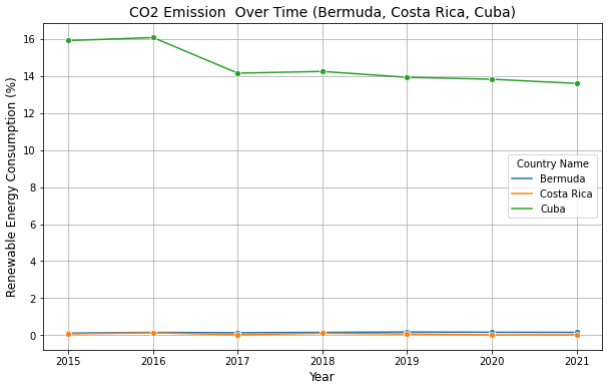


Fig. 4. CO2 Emission Graph for Bermuda, Cuba, and Costa Rica.

B. *Correlation Analysis*

1) ***Bermuda****:* Renewable Energy Consumption: Minimal, starting at 0.6 percent in 2015, peaking at 1.1 percent in 2016, and gradually increasing after 2018. CO2 Emissions: Rise from 0.12 metric tons per capita in 2015 to a peak of 0.17 in 2019, followed by a gradual decline through 2021. Correlation: The low renewable energy share correlates with higher CO2 emissions. As renewable energy slightly increased after 2018, CO2 emissions began to decline, showing a weak inverse relationship.

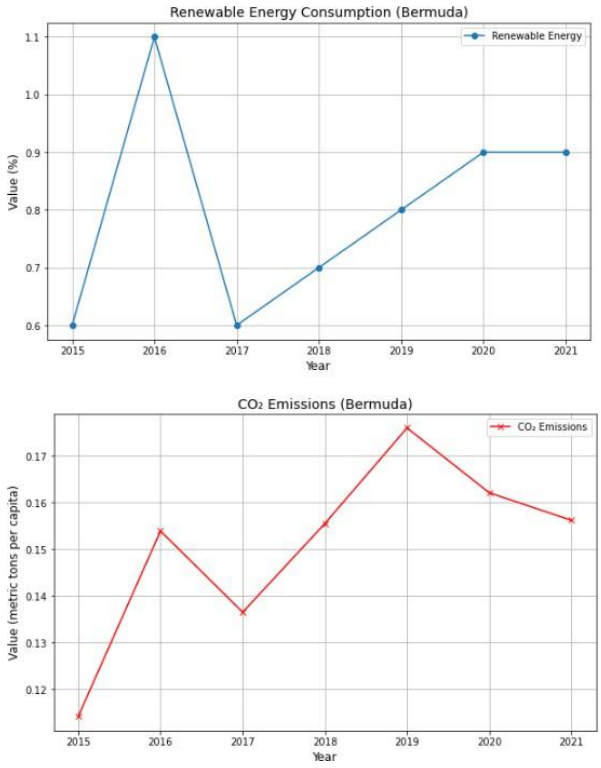


Fig. 5. Correlation of Renewable energy Consumption and Co2 Emissions Graph for Bermuda

2) ***Costa Rica****:* Renewable energy consumption is generally high, with a decline from 38 percent in 2015 to around 33 percent in 2018, followed by an increase in 2020 and another drop in 2021. CO2 Emissions: fluctuate slightly but remain very low (0.02–0.12 metric tons per capita), with a notable drop in 2020. Correlation: As renewable energy increased in 2020, CO2 emissions decreased significantly. The decline in renewable energy in 2021 corresponds to the lowest emissions due to Costa Rica's already minimal reliance on fossil fuels.

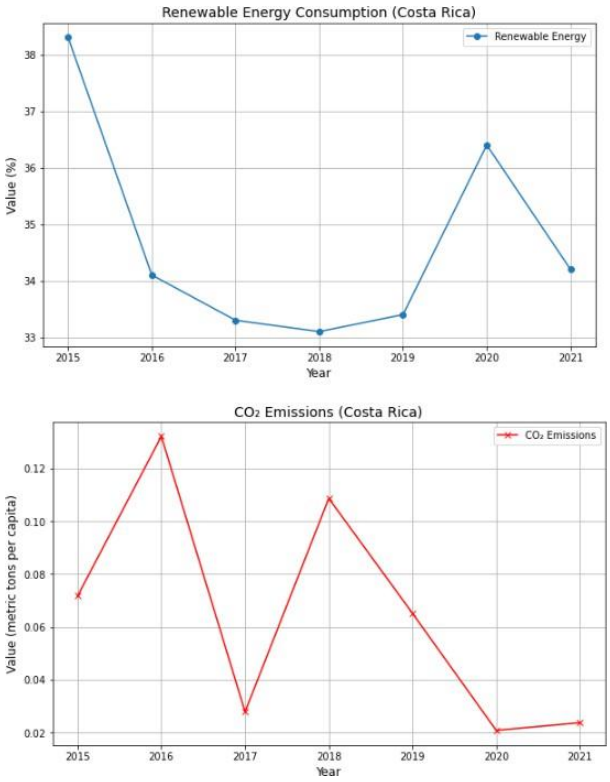


Fig. 6. Correlation of Renewable Energy Consumption and CO2 Emissions Graph for Costa Rica.

3) ***Cuba****:* Renewable Energy Consumption shows fluctuations, with an increase from 2015 to 2020 peaking at 25 percent, followed by a sharp decline in 2021. CO2 Emissions: Decrease steadily from 16 metric tons per capita in 2015 to around 13.5 in 2021. Correlation: When renewable energy consumption increased (2015-2020), CO2 emissions consistently decreased. However, the sharp drop in renewable energy in 2021 did not immediately affect the downward trend in CO2 emissions, likely due to other contributing factors.

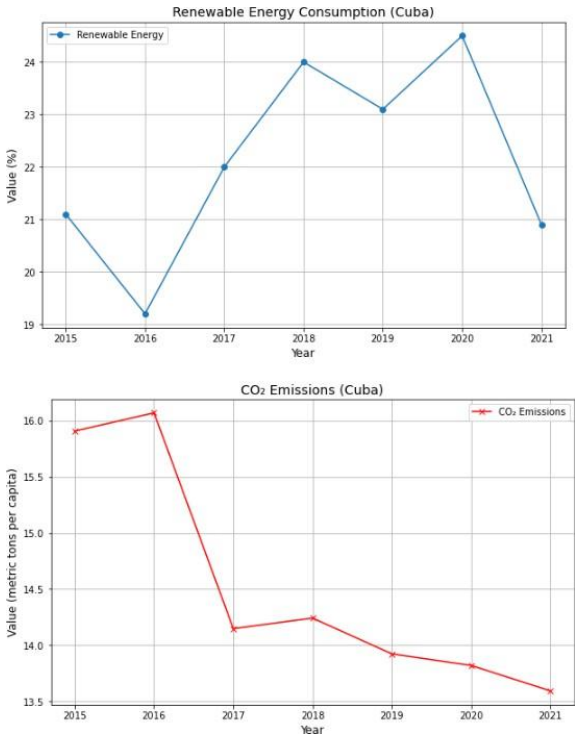


Fig. 7. Correlation of Renewable Energy Consumption and CO2 Emissions Graph for Cuba.

Does consumption of renewable energy sources decreases the CO2 emission? Yes, the analysis supports the claim that increased consumption of renewable energy sources decreases CO2 emissions. The extent of this effect varies by country but is consistently observed across the analyzed data. Countries with a higher share of renewable energy, such as Costa Rica, exhibit the strongest reductions in CO2 emissions, while minimal renewable adoption, as seen in Bermuda, results in weaker correlations. Across the three countries, the data indicates that increased renewable energy consumption generally correlates with decreased CO2 emissions. The strength of this relationship depends on: The share of renewable energy in the energy mix. Other factors influencing emissions include energy efficiency, industrial activity, and fossil fuel consumption.

IV. Conclusion

Renewable Energy and CO2 Emission Relationship:

There is a negative correlation between renewable energy consumption and CO2 emissions in most cases. An increase in renewable energy share in the energy mix is associated with a decline in CO2 emissions, as seen in countries like Cuba and Costa Rica. Countries with a high share of renewable energy (e.g., Costa Rica) exhibit significantly lower CO2 emissions, supporting the idea that transitioning to renewable energy is effective in reducing greenhouse gas emissions. For countries with minimal renewable energy usage (e.g., Bermuda), the impact on CO2 emissions is less pronounced, highlighting the importance of substantial renewable adoption for meaningful emission reductions.

Regional Variability:

The relationship varies across countries due to differences in energy infrastructure, reliance on fossil fuels, economic activities, and government policies. For example:

Costa Rica: High renewable energy use correlates with consistently low CO2 emissions, showcasing the effectiveness of renewable adoption.

Cuba: Despite fluctuations in renewable energy consumption, CO2 emissions steadily decrease, suggesting additional factors (e.g., energy efficiency measures, reduced fossil fuel use) contribute to emissions reductions.

Bermuda: Limited renewable energy adoption correlates with higher CO2 emissions, demonstrating the challenges faced by countries with minimal renewable infrastructure.

V. Limitation and Further Study

Limitation

While the analysis establishes a general correlation between renewable energy consumption and CO2 emissions, several limitations must be acknowledged:

- The data is annual and lacks regional or seasonal details, potentially oversimplifying the relationship between renewable energy and CO2 emissions.
- Renewable energy consumption percentages do not provide insights into the types of renewable sources used (e.g., hydro, solar, wind), which might vary in their impact on emissions.
- Study the specific types of renewable energy and their distinct impacts on emissions.
- CO2 emissions data captures only direct emissions and does not account for lifecycle emissions from renewable energy infrastructure (e.g., manufacturing solar panels).
- The analysis focuses solely on CO2 emissions, ignoring other greenhouse gases such as methane (CH4) or nitrous oxide (N2O), which could also play a role in total emissions.

Future Studies

- Include additional variables like industrial energy use, population growth, and economic activity.
- Analyze data with higher temporal resolution (e.g., monthly data) to capture short-term trends and seasonal effects.
- Study the specific types of renewable energy and their distinct impacts on emissions.
- Investigate the role of government policies and incentives in promoting renewable energy adoption and reducing CO2 emissions. Broaden the scope to include other greenhouse gases for a more holistic view of emissions reductions.